American National Standard for Calibration —

Requirements for the Calibration of Measuring and Test Equipment

ANSI/NCSL Z540.3-2006 (R2013)

Approved American National Standard

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for Calibration —

Requirements for the Calibration of Measuring and Test Equipment

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Foreword

The objective of this National Standard is to establish the technical requirements for the calibration of measuring and test equipment. This is done through the use of a system of functional components. Collectively, these components are used to manage and assure that the accuracy and reliability of the measuring and test equipment are in accordance with identified performance requirements.

In implementing its objective, this National Standard describes the technical requirements for establishing and maintaining:

– the acceptability of the performance of measuring and test equipment;
– the suitability of a calibration for its intended application;
– the compatibility of measurements with the National Measurement System; and
– the traceability of measurement results to the International System of Units (SI).

In the development of this National Standard attention has been given to:

– expressing the technical requirements for a calibration system supporting both government and industry needs;
– applying best practices and experience with related national, international, industry, and government standards; and
– balancing the needs and interests of all stakeholders.

In addition, this National Standard includes and updates the relevant calibration system requirements for measuring and test equipment described by the previous standards, Part II of ANSI/NCSL Z540.1 (R2002) and Military Standard 45662A.

This National Standard is written for both Supplier and Customer, each term being interpreted in the broadest sense. The “Supplier” may be a producer, distributor, vendor, or a provider of a product, service, or information. The “Customer” may be a consumer, client, end-user, retailer, or purchaser that receives a product or service.

Reference to this National Standard may be made by:

– customers when specifying products (including services) required;
– suppliers when specifying products offered;
– legislative or regulatory bodies;
– agencies or organizations as a contractual condition for procurement; and
– assessment organizations in the audit, certification, and other evaluations of calibration systems and their components.

This National Standard is specific to calibration systems. A calibration system operating in full compliance with this National Standard promotes confidence and facilitates management of the risks associated with measurements, tests, and calibrations.

This National Standard was approved for submittal to the American National Standards Institute (ANSI) by the Accredited Standards Committee on General Requirements for Calibration Laboratories and Measuring and Test Equipment, Z540. Committee approval of this standard does not necessarily imply that all committee members voted for its approval.

At the time it approved this National Standard, the Committee had the following members. This National Standard was prepared by Working Group One, who are indicated by *.

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American National Standard for Calibration —

Requirements for the Calibration of Measuring and Test Equipment

1 Scope

1.1 This National Standard prescribes requirements for a calibration system to control the accuracy of the measuring and test equipment used to ensure that products and services comply with prescribed requirements.

1.2 Compliance with regulatory and safety requirements on the operation of the calibration system is not covered by this National Standard.

1.3 The notes given provide clarification of the text, examples and guidance. They do not contain requirements and do not form an integral part of this National Standard.

1.4 This National Standard and any procedure or document executed in implementation thereof shall be in addition to and not weaken or detract from other contract requirements.

2 References

2.1 Normative

The following documents contain provisions which, through reference in this text, constitute provisions of this National Standard.

VIM, *International vocabulary of basic and general terms in metrology*, issued by BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, and OIML, 1993.

2.2 Informative

The following supplemental documents are referred to in the text of this National Standard and apply only to the extent described.

ANSI/NCSL Z540-2-1997 (R2002), *U.S. Guide to the Expression of Uncertainty in Measurement.*

ANS/ISO/IEC 17025:2005, *General requirements for the competence of testing and calibration laboratories*

NCSLI RP-1-1996, *Establishment and Adjustment of Calibration Intervals.*

3 Terms and definitions

For the purposes of this National Standard, the relevant terms and definitions given in this clause and the VIM shall apply, and in this order of authority.

3.1 Calibration
The set of operations that establish, under specified conditions, the relationship between values of quantities indicated by a measuring instrument or measuring system, or values represented by a material measure or a reference material, and the corresponding values realized by standards.

> NOTE: As used herein, a calibration may also include verification that specified requirements, such as tolerances, performance, or other decision criteria, have been met. It may also include actions to take when requirements have not been met.

3.2 Calibration system, calibration program
The set of interrelated or interacting elements necessary to maintain the measurement performance of measuring and test equipment to defined requirements.

3.3 Guard band
The offset of a measurement decision value from a stated specification or tolerance.

3.4 Measurement assurance
The result of a process to provide adequate confidence that a measurement will satisfy stated requirements.

3.5 Measurement decision risk
The probability that an incorrect decision will result from a measurement.

3.6 Measurement process
The set of operations to determine the value of a measurement quantity.

3.7 Measurement quantity
An attribute of a phenomenon, artifact, or substance that may be distinguished qualitatively and determined quantitatively.

> NOTE: Attributes include characteristics of an object or unit under calibration.

3.8 Measurement reliability
The probability that all the applicable measurement quantities of measuring and test equipment are within tolerance.

> NOTE: Measurement reliability may be considered to be a function of time and described at some point in time or over some interval of time.

3.9 Measuring and test equipment
The measuring instrument, measurement standard, reference material or auxiliary apparatus, or a combination thereof, necessary to realize a measurement process.

> NOTE: This also includes individual items of equipment and measuring systems comprised of several items that are used in calibration, test, inspection, and verification.

3.10 Organization
A group of people and facilities with an arrangement of responsibilities, authorities, and relationships.

> NOTE 1: An organization can be governmental or private.

> NOTE 2: In some instances, an organization may consist of one person.

> NOTE 3: An organization can be a company, corporation, firm, enterprise, institution, association, or a part or combination of parts of these.

3.11 Test uncertainty ratio
The ratio of the span of the tolerance of a measurement quantity subject to calibration, to twice the 95% expanded uncertainty of the measurement process used for calibration.

> NOTE: This applies to two-sided tolerances.

3.12 Tolerance
Extreme values of an error permitted by specifications, regulations, etc., for a given measuring instrument, test, or measurement application.

3.13 Validation
Confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled.

4 General requirements

4.1 Calibration system objectives

The organization shall establish, document, operate, and improve a system to manage the calibration of measuring and test equipment. The organization shall identify and include measuring and test equipment in the calibration system having an influence on the quality of the organization’s product and its conformity to determined requirements.

Requirements for the product include:

– requirements specified by the customer, including the requirements for delivery and post-delivery activities;
– requirements not stated by the customer but necessary for specified or intended use, where known;
– statutory and regulatory requirements related to the product; and
– any additional requirements determined by the organization.

The organization shall ensure that:

a) performance requirements for measuring and test equipment are established;
b) the calibration system meets the performance requirements; and
c) customer requirements are met.

Performance requirements for measuring and test equipment shall be derived from the application of the equipment, including the measurement process, test method, or calibration procedure.

> NOTE: Requirements may be expressed as: measurement parameter range,

resolution, and stability; measurement uncertainty and measurement reliability; accuracy or tolerances; operating range for various influence characteristics, such as temperature and humidity; and other factors affecting the suitability of the equipment for the application requirements.

The organization shall ensure the availability of necessary resources and address reporting and accountability requirements for the calibration system. The reporting and accountability requirements shall be such that potential conflicts of interests do not adversely influence compliance with the requirements of this National Standard.

4.2 Quality objectives

The organization shall define and establish measurable quality objectives for the calibration system and its components.

> NOTE: A component of a calibration system includes: a separate or distinct function or activity; a functional entity or organization; or another accountable element of the calibration system.

The organization shall document policies, systems, programs, procedures and instructions to the extent necessary to assure the quality of the measuring and test equipment performance. The documentation shall include the following:

a) objectives of the calibration system;
b) performance criteria for the calibration system;
c) requirement for applicable personnel to understand the operation of the calibration system and to implement the policies and procedures in their work;
d) reference to calibration system requirements, including supporting procedures and instructions;
e) the roles, responsibilities, and organizational location of the calibration system components; and
f) procedures for authorizing activities where such authorization is required by this National Standard.

4.3 Personnel requirements

4.3.1 Responsibilities of personnel

The organization shall document the responsibilities of its personnel assigned to or utilizing the resources of the calibration system.

4.3.2 Competence and training

Personnel involved in the calibration system shall be competent and have demonstrated the ability to perform their assigned tasks.

> NOTE: Competence may be achieved through education, training, and experience, and may be demonstrated by testing or observed performance.

Training shall be provided to address identified needs. Records of training activities shall be maintained. The effectiveness of the training shall be evaluated and recorded.

Personnel shall be made aware of the extent of their responsibilities and accountabilities, and their impact on the calibration system and product quality.

4.4 Information resources

4.4.1 Procedures, software, and documentation

Calibration system processes shall be documented and validated to ensure proper implementation, consistency of application, and the validity of results.

Calibration system documentation shall identify the authorized calibration procedures and calibration intervals for measuring and test equipment included in the calibration system.

All procedures, software, and documentation, including changes, shall be authorized, controlled, current, and available.

Deviation from calibration system processes, software, and documentation shall occur only if the deviation has been documented, technically justified, and authorized.

4.4.2 Records and data control

Records containing information required for the operation and assessment of the calibration system shall be maintained. Procedures for record and data control shall be documented, used, and ensure the identification, storage, protection, confidentially, retrieval, retention time, and disposition of records and data.

Records shall include: performance and calibration requirements for measuring and test equipment; results of calibration and other measurements; measurement reliability and/or uncertainty information; nonconformance data; customer feedback; and, personnel qualifications.

Calibration system data shall be controlled to the extent necessary to maintain the integrity and confidentiality of the data.

4.5 Shipping and handling

The organization shall establish, implement, and maintain documented procedures for shipping, receiving, handling, transporting, storing and dispatching measuring and test equipment, in order to prevent abuse, misuse, damage, changes to performance, or any adverse effect on the calibration or condition of the equipment.

The procedures shall include provisions necessary to protect the integrity of the calibration of measuring and test equipment and to protect the interests of the organization and the customer and/or suppliers.

5 Calibration system implementation

5.1 Calibration requirements

The organization shall include all measuring and test equipment in the calibration system that have an influence on the quality of the organization's product.

The organization shall document and use procedures for the determination of measuring and test equipment to be included in the calibration system.

The calibration requirements determination procedures shall:

a) ensure that the performance requirements of the measuring and test equipment are established and based on their intended application;

   NOTE: The performance requirements of the measuring and test equipment may be a composite requirement based on multiple applications or those represented by an equipment manufacturer's specifications.

b) ensure that the calibration requirements of the measuring and test equipment are established and meet their performance requirements;
c) include a description of the process for adding or removing measuring and test equipment from the calibration system; and
d) provide for review of the performance requirements of the measuring and test equipment at designated intervals to assure calibration requirements remain suitable for the intended application.

5.2 Measuring and test equipment

5.2.1 Identification

Measuring and test equipment included in the calibration system shall be clearly identified, individually or collectively, as in the case of a measurement system. This equipment shall be distinguishable from other equipment.

The calibration status of the measuring and test equipment included in the calibration system shall be evident to the user. The calibration status shall include the following information:

a) unambiguous identification of the measuring and test equipment;
b) date of calibration;
c) date due for next calibration;
d) conformance to specifications or limitations of use; and
e) calibration-servicing component.

The calibration status may be indicated by a label and/or tag applied directly to the measuring and test equipment, or companion case, or by some other means clearly evident to the user.

5.2.2 Adjustment control

Access to means of adjustment that modify the calibrated performance of measuring and test equipment shall be controlled to prevent and detect unauthorized use. Controls may include seals or other types of safeguards. The requirement does not apply to adjustment means that are intended to be set by the user without the need for external references; for example, zero adjusters.

5.2.3 Nonconforming measuring and test equipment

Measuring and test equipment that are included in the calibration system and suspected or known:

a) to have been damaged, overloaded, or mishandled;
b) to have malfunctioned in such a way that may invalidate its intended use;
c) to produce incorrect measurement results;
d) to have been used beyond its designated calibration interval without an authorized temporary extension;
e) to have damaged, broken, bypassed, or missing seals or adjustment access controls; or
f) to have been exposed to influencing quantities that can adversely affect its intended use;

shall be removed from service and identified by prominent labeling or marking. Such equipment shall not be returned to service until the reasons for its nonconformity have been investigated and resolved, and it has been recalibrated.

5.2.3.1 As-found calibration

Criteria shall be established to notify the using organization when measuring and test equipment are found not to meet measurement performance requirements during calibration. Calibration results shall be monitored and if a criterion is exceeded, the using organization shall be notified of the nonconformity in a timely manner. The notification shall include sufficient calibration data to evaluate the effect of the non-conformity on the organization's products.

> NOTE: The criteria may be such as to require all the out-of-tolerance measuring and test equipment to be reported or only those that have exceeded the tolerance by a determined amount.

5.2.3.2 Use of nonconforming measuring and test equipment

Nonconforming measuring and test equipment, which are not returned to their intended performance requirements, shall be clearly marked or otherwise identified. Procedures for use of such equipment shall ensure that the altered status is clearly apparent and includes identification of any limitations of use.

5.3 Calibration of measuring and test equipment

Calibration of measuring and test equipment shall be in accordance with the requirements of this National Standard. Calibration may be performed within or outside a designated calibration facility, e.g., in situ, on-site, or at a customers facility, provided compliance with the requirements of this National Standard is maintained. The scope of the calibration capability shall be consistent with the calibration requirements and provide levels of measurement decision risk acceptable to both the customer and supplier.

a) Where calibrations provide for reporting measured values, the measurement uncertainty shall be acceptable to the customer and shall be documented.
b) Where calibrations provide for verification that measurement quantities are within specified tolerances, the probability that incorrect acceptance decisions (false accept) will result from calibration tests shall not exceed 2% and shall be documented. Where it is not practicable to estimate this probability, the test uncertainty ratio shall be equal to or greater than 4:1.

> NOTE: Achieving these requirements may involve adjustment and management of calibration system parameters such as: measurement reliability, calibration intervals, measurement uncertainty, calibration tolerances, and/or guard bands.

Calibration-servicing components may be considered competent to provide calibration services when they have been accredited to meet ANS/ISO/IEC 17025, including the requirements of this sub-clause, or otherwise found to be in conformance by an authority acceptable to the customer. Competence shall be confirmed for the required calibration services and reflected in a scope of accreditation or in a similar listing of calibration capability and conformity in the non-accredited laboratory.

All measuring and test equipment included in the calibration system, including measuring systems, calibration equipment, reference standards and material, and other inspection and monitoring equipment, shall be calibrated prior to use and recalibrated at predetermined intervals to ensure acceptable measurement uncertainty, traceability, and reliability. Intervals may be based on usage or time since last calibration.

When there is doubt as to the suitability of an item for calibration, when an item does not conform to the description provided, or the calibration required is not specified in sufficient detail, the calibration-servicing component shall consult with the customer for further instructions before proceeding and shall record the discussion.

5.3.1 Calibration procedures

Calibrations shall be performed using calibration procedures that:

– address the measuring and test equipment performance requirements;
– are acceptable to the customer;
– are current and appropriate for the calibrations; and
– provide reasonable assurance that the calibration results are as described.

All calibration procedures shall:

a) contain sufficient information on requirements for the associated measurements and instructions to perform the calibrations. In addition, the number of different measurement quantities and values in a calibration procedure shall be sufficient to ensure conformity of the measuring and test equipment to determined requirements;
b) provide for determining and recording the as-found performance of the measuring and test equipment being calibrated; and
c) be suitable for use by the calibration staff to perform and reproduce a calibration to the established performance criteria. Deviation from calibration procedures shall occur only if the deviation has been documented, technically justified, authorized, and accepted by the customer.

NOTE: Calibration procedures may be unique to a specific application or general in nature, applying to multiple applications.

When a calibration procedure proposed by the customer is considered to be inappropriate, the customer shall be notified.

5.3.1.1 Contents

Calibration procedures shall include the following information:

a) identification and document control information;
b) scope and/or description of item to be calibrated;
c) measurement quantities and ranges to be determined for the item to be calibrated and any associated tolerances;
d) minimum performance requirements of the equipment to be used for calibration, including measurement and reference standards, and reference materials;
e) environmental conditions required and any stabilization period needed;
f) description of steps associated with the calibrations to be performed;
g) criteria and/or requirements for calibration decisions, such as approval or rejection; and
h) data to be recorded and method of analysis and presentation.

NOTE: Calibration procedures may include the contents of standards, methods and/or procedures supplemented with additional details to ensure consistent application.

5.3.1.2 Validation

Calibration procedures and their modifications, shall be validated. The validation shall be as extensive as is necessary to meet the needs of the procedure's application.

5.3.2 Measurement assurance procedures

Measuring processes incorporating measurement assurance methods, such as statistical process control, shall use a measurement assurance procedure. This procedure shall be systematically applied and include stated measurement uncertainty or reliability goals, control criteria, and methodology to verify that the goals and criteria are being attained. The controls shall be adjusted when the results of the previous measurements indicate that such action is appropriate to maintain acceptable measurement uncertainty or reliability. Measurement assurance controls may be based on the use of calibrated check standards, usage, and/or time since the last performance. The measurement assurance procedure shall include mandatory instructions to preclude the use of the measuring process that exceeds its controls.
The measurement assurance procedure and any associated measuring and test equipment shall be documented as a calibration procedure in

accordance with the provisions of this National Standard.

5.3.3 Measurement uncertainty and traceability

The uncertainty and traceability of all measurement results associated with processes included in the calibration system shall meet the requirements of their applications. Measurement uncertainty components which have an influence on such measurement results shall be included in the estimates of measurement uncertainty.

5.3.3.1 Expression of measurement uncertainty

A documented procedure shall be used to estimate and express the uncertainty of measurement for all calibrations. As a minimum, the procedure shall address:

a) sources of measurement uncertainty;
b) estimation and combining of uncertainties;
c) conditions and assumptions;
d) documentation and reporting criteria; and
e) bibliography.

NOTE 1: Sources of measurement uncertainty that should be addressed include:
- measuring and test equipment associated with the calibration;
- measurement and calibration methods and procedures;
- measurement traceability;
- measurement repeatability and reproducibility;
- calibration quality monitoring;
- changes over time; and
- influence quantities.

NOTE 2: For guidance on the expression of measurement uncertainty, see ANSI/NCSL Z540-2-1997 (R2002).

5.3.3.2 Measurement traceability

The results of a calibration or measurement shall be traceable through a controlled, unbroken chain of competent calibrations to and through the National Institute of Standards and Technology to the SI units of measurement.

This traceability to a national measurement institute other than the National Institute of Standards and Technology is acceptable when:

a) a mutual recognition agreement, such as the Comité International des Poids et Mesures (CIPM) Mutual Recognition Arrangement (MRA), is in effect with the National Institute of Standards and Technology and sufficient equivalence of applicable calibration services exists; or
b) when the calibration service of the National Institute of Standards and Technology is not available or does not meet the measurement performance requirements.

Where traceability to SI units through national metrology institutes is not available, or SI units are not established, a consensus standard including a reference standard and related calibration procedures, which are clearly specified and mutually agreed upon by all parties concerned, shall be applied.

NOTE 1: Calibration-servicing components fulfilling the requirements of this National Standard may be considered competent. Objective evidence of assessment that includes the requirements of this National Standard and are applicable to the customer's requirements may also be considered.

NOTE 2: The unbroken chain of calibrations or comparisons may be achieved in several steps carried out by different competent calibration-servicing components.

NOTE 3: Reference materials provided or designated by a national metrology institute may be considered as a reference standard.

5.3.4 Calibration equipment

All measuring and test equipment required for the correct performance of calibrations and related measurements, including calibration and reference standards and reference material, shall be available to the calibration-servicing component. In those cases where the calibration-servicing component needs to use equipment outside its permanent control, it shall ensure that the requirements of this National Standard are met.

Measuring and test equipment used for calibration and associated measurements shall be capable of achieving the performance required and shall comply with specifications relevant to the calibrations performed.

Measuring and test equipment that may affect the results of the calibrations shall be calibrated and included in a calibration system meeting the requirements of this National Standard.

5.3.4.1 Documentation
Instructions on the use and operation of all relevant equipment shall be available to the calibration-servicing component where the absence of such instructions could jeopardize the results of calibrations.

5.3.5 Calibration personnel
Personnel who operate specific equipment, perform calibrations or measurements, evaluate results, give opinions and interpretations, prepare calibration records, and/or sign calibration reports shall be competent. When using personnel who are undergoing training, appropriate supervision shall be provided.

5.3.5.1 Authorization
Specific personnel shall be authorized: to perform particular types of calibrations; to issue calibration procedures and reports; to give opinions and interpretations; to prepare calibration records; and to operate particular types of equipment. Records of such authorization shall be maintained, readily available, and include the date of authorization.

5.3.6 Influence factors and conditions
All factors and conditions of the calibration area that adversely influence the calibration results shall be defined, monitored, recorded, and mitigated to meet calibration process requirements.

> NOTE: Influencing factors and conditions may include temperature, humidity, vibration, electromagnetic interference, dust, etc.

Calibration shall be stopped when the adverse effects of the influence factors and conditions jeopardize the results of the calibration.

5.3.7 Calibration quality
Calibration quality shall be monitored to assure the validity of the calibration services. The monitoring process shall be documented and the resulting data recorded and evaluated. This monitoring process may include, but not be limited to, the following:

a) sampling of calibrated measuring and test equipment;
b) calibrations using different procedures or methods;
c) regular use of measurement assurance methods; and/or
d) participation in measurement inter-comparison programs.

When the results of any of the procedures indicate suspect calibrations, appropriate actions shall be taken to identify the root cause and take any corrective and preventive actions.

5.3.8 Reporting the results
The results of each calibration or series of calibrations shall be reported accurately, clearly, unambiguously, and objectively.

The results reported shall include all information requested by the customer and necessary for the interpretation of the calibration results, and all information required by the calibration procedure used.

5.3.8.1 Calibration reporting
Where a calibration report is issued, the report shall include at least the following information:

a) title;
b) identification of the calibration-servicing component, and the location where the calibrations were carried out, if different;
c) unique identification of the calibration report;
d) identification of the customer;
e) identification of the calibration procedure used;
f) description of, the condition of, and unambiguous identification of the item calibrated;
g) date of performance of the calibration;
h) calibration results with, where appropriate, the units of measurement;
i) calibration results before and after adjustment or repair, if available;
j) person authorizing the calibration report;
k) statement of measurement traceability;
l) factors and conditions under which the calibrations were made that have an influence on the measurement results; and
m) uncertainty of measurement, where appropriate, and/or a statement of conformance.

Where a statement of conformance with specified calibration performance requirements and/or calibration procedure is provided, the extent of conformance shall be provided and included in the calibration records.

5.3.8.2 Opinions and interpretations

When opinions and interpretations are included, the basis upon which the opinions and interpretations have been made shall be documented. Opinions and interpretations shall be clearly marked as such in a calibration report.

5.3.8.3 Control and source

When it is necessary to amend or issue a complete new calibration report, it shall be uniquely identified and shall contain a reference to the original that it amends or replaces.

5.3.9 Calibration records

Records of the calibration process shall be completed to attest to the correctness of the results, as appropriate. These records shall be maintained and made available.
Records of the calibration process shall demonstrate whether measuring and test equipment have satisfied the performance requirements specified. The records should be sufficient to demonstrate the traceability of the measurements and reproduce the calibration results under conditions close to the original conditions.

Records shall be maintained of each item of equipment and software where its use contributes to the results of the calibrations performed.

The records shall include the following:

a) description and unique identification of the measuring and test equipment (manufacturer, model, nomenclature, serial number, etc.);
b) identification of the calibration-servicing component;
c) identification of the calibration procedure used;
d) date on which the calibration was completed;
e) as-found measurement performance condition of the equipment;
f) calibration actions taken (adjusted, repaired, new value assigned, limited, derated, modified, etc.);
g) calibration results and the as-returned measurement performance condition of the equipment;
h) limitations of use;
i) assigned calibration interval; and
j) identification of calibration reports.

Generation, amendment, issuance, and deletion of records shall be authorized. In addition, the reason for an amendment or deletion of a record shall be documented.

5.4 Calibration intervals

Calibration intervals and/or process controls shall be established for measuring and test equipment that are included in the calibration system to monitor and maintain equipment performance to the stated application requirements.

The procedures for establishing and adjusting calibration intervals and process controls shall be documented and utilized.

The calibration and other measurement process data shall be regularly monitored to evaluate the achievement of the performance requirements. The calibration intervals shall be adjusted to maintain the measurement performance requirements.

5.4.1 Measuring and test equipment calibration intervals

Measuring and test equipment within the scope of the calibration system shall be calibrated at periodic intervals established and maintained to assure acceptable measurement uncertainty, traceability, and reliability.

The calibration interval process shall be free from conflicts of interest and systematically applied. The process shall be based on data resulting from the calibration of measuring and test equipment. The process shall also include stated measurement reliability or measurement uncertainty requirements and a method of verifying that the requirements are being attained.

Calibration intervals shall be reviewed regularly and adjusted when necessary to assure continuous compliance of the specified measuring and test equipment performance requirements.

The calibration system shall include mandatory recall of measuring and test equipment to assure timely recalibrations and preclude use of an item beyond its calibration due date. All exemptions from periodic calibration shall be documented and authorized. The recall system may provide for the temporary extension of the calibration due date for limited periods of time under specified conditions, such as completion of a test or calibration that is in progress, that do not unreasonably impair the satisfaction of the customer's requirements.

> NOTE: NCSLI RP-1-1996 should be used as a basis for the analytical process in the determination of calibration intervals.

5.4.2 Measuring process control intervals

Measuring processes incorporating measurement assurance methods, such as statistical measurement process control, that are included in the calibration system shall incorporate provisions to perform the related procedure at determined times, events, or intervals. Such determinations shall be established and maintained to assure acceptable measurement uncertainty, traceability, and reliability.

The measuring and test equipment system used in the measurement assurance processes shall be calibrated at intervals established and maintained to assure acceptable measurement uncertainty, traceability, and reliability.

5.5 Outside suppliers

The management of the calibration system shall define and document the requirements for products and services to be provided by outside suppliers when they have an effect on the calibration system.

Where the outside supplier performs as a component or a supplemental component of the calibration system, the applicable requirements of this National Standard shall be applied.

These suppliers shall be evaluated and selected on their ability to meet the applicable requirements of this National Standard.

> NOTE: Assessment of a supplier by a third party that is acceptable to the customer, and includes the requirements of this National Standard, may serve as a basis for determining the supplier's compliance to these requirements.

Criteria for selection, monitoring and evaluation shall be defined and documented, and the evaluation results shall be recorded. Records shall be maintained of the products or services provided by outside suppliers.

6 Calibration system assessment and improvement

The organization shall plan and implement an assessment process including, review, auditing, monitoring, analysis, and improvement:

a) to ensure conformity of the calibration system with this National Standard;
b) to continually improve the calibration system; and
c) to determine the suitability and effectiveness of the calibration system.

> NOTE: Assessment of a calibration system component by a third party that is acceptable to the customer and includes the requirements of this National Standard, may serve as a basis for determining the component's compliance to these requirements.

Customer feedback information related to satisfaction and achievement of requirements shall be documented and monitored.

Continual improvement of the calibration system shall be a planned and managed activity, and based on the results of the audits, management reviews and of other relevant factors, such as feedback from customers and functional components. Management shall also review and identify potential opportunities to improve the calibration system, and modify it as necessary.

Personnel involved with the assessment of the calibration system, including review, audit, certification, accreditation, etc., shall be competent and have demonstrated the ability to perform their assigned tasks.

6.1 Management review

The organization shall ensure the systematic review of the calibration system at planned intervals to ensure the continual adequacy, effectiveness, and suitability of the system to achieve established objectives. The results of all reviews and actions taken shall be recorded.

6.2 Calibration system audit

The organization shall establish and document criteria and procedures for auditing the calibration system. The organization shall ensure the systematic and independent audit of the calibration system and objectively evaluate the results to determine the extent to which the audit criteria are fulfilled.

The results of all audits of the calibration system, and all changes to the system, shall be recorded and used.

6.3 Calibration system monitoring

Within the processes comprising the calibration system, the following shall be monitored at established intervals:

a) performance quality of measuring and test equipment;
b) calibration interval suitability; and
c) calibration service quality and traceability.

Monitoring shall be in accordance with documented procedures. This shall include determination of applicable methods, including statistical techniques, and the extent of their use.

Monitoring of the calibration system shall provide for the prevention of deviations from requirements by ensuring the prompt detection of deficiencies and timely actions for their correction. This monitoring shall be commensurate with the risk of failure to comply with the specified requirements.

The results of monitoring and any resulting corrective actions shall be recorded.

6.4 Customer assessment, verification, and feedback

All operations of the calibration system shall be subject to assessment and verification by the customer. Customer feedback information related to satisfaction and achievement of requirements shall be collected, documented, evaluated, and used.

6.5 Corrective and preventive action

When a relevant calibration system functional component does not meet the specified requirements, or when relevant data shows an unacceptable pattern, action shall be taken to identify the cause and eliminate the discrepancy. Also, action shall be taken to determine if preventive measures are appropriate to preclude occurrence of other discrepancies in another area of the calibration system.

Corrective and preventive actions shall be verified, documented, and implemented.